

Date	15/07/2026
Team ID	6a3a4b3c8fd3a72620012b1d
Project Name	Smart Lender (Loan Approval Prediction)
Maximum Marks	4 Marks

Solution Requirements Template

Functional and Non-Functional requirements for the Smart Lender Prediction System.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Description	Priority
1	User Form Input	Web interface accepting applicant personal & financial parameters.	High
2	Data Preprocessing	Automatic categorical mapping and numerical scaling using StandardScaler.	High
3	Model Inference	Run Random Forest Classifier on scaled features to predict approval.	High
4	Result Presentation	Render a simple, clear results page showing approval status.	High
5	Error Handling	Validate inputs to prevent system crashes.	High

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Low latency response times for predictions.	< 1.5 seconds per prediction
2	Model Accuracy	Accurate credit assessment.	> 80% test set accuracy
3	Security	Input validation and error handling.	Sanitized inputs, no crash on bad types
4	Usability	Responsive UI working across multiple screens.	User friendly CSS layout